

# Social Network Analysis

From pre-teen to young-adult: The  
(potentially gendered) development  
& maintenance of emotional  
support-networks

Paula Hensums





## Seemingly same but actually different: A comparative study of class friendship networks and support-seeking networks

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### Abstract

Peer support in school is vital for students' mental health, particularly within China's distinctive educational context centered on stable, fixed classes. Class friendship networks and support-seeking networks are widely studied interpersonal networks in peer support research. Previous studies have rarely conducted in-depth comparisons between these two types of networks, leading to confusion in their usage, which impedes the further development of school peer support research. This cross-sectional study examines a sample of 799 second-year junior high school students across 29 classes (48.6% female) from China, exploring the connections and differences between the two networks by comparing their structures and functions through social network analysis. The results indicate the following: (1) The two networks exhibit partial overlap, though there are differences in structural indicators at individual, dyadic and triadic, and group level; (2) While the two networks share similar key configurations, the role of these configurations in network formation differs; (3) Perceived peer support exerts a greater influence on the formation of support-seeking networks; (4) Both networks are closely associated with mental health, but support-seeking networks show stronger associations with well-being than friendship networks. The findings suggest that although the two networks are somewhat interconnected, they differ in structure, formation processes, and their impact on students, indicating they are distinct types of networks within class-based peer environments. Therefore, they should be differentiated in future research and applications.

**Keywords** Class peer support · Exponential random graph models · Friendship networks · Support-seeking networks · Social network analysis

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*“Seemingly same but actually different:  
A comparative study of class friendship  
networks and support-seeking networks”  
(Wang, Jiang, Ye & Bian, 2025)*

Comparison between network types

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# Article 1: Comparison between networks

- **Friendship Networks (FNs) & Support Networks (SNs)**

- FNs: Being friends (assistance with daily life/sense of belonging)
  - *"Do you have any good friends in your class, who are they?"*

- SNs: Asking for help (academic/emotional problems)
  - *"When you have worries or encounter problems, would you seek help from classmates? If so, write down their student ID's?"*

- Focus on connections and differences in **structures** (overlap, size, transitivity, and density) & **functions** (mental health: internalizing problems & well-being)
- Chinese students in Beijing (29 classes)
- Cross-sectional in 2016

# Article 1: Analyses

- Excel for potential overlap between FNs & SNs
  - Relationship status of all dyads in each class (via two adjacency matrices)
  - Multiply the two matrices, check for overlapping ties
  - Count the total number of overlapping ties
- Use R packages *sna* and *igraph* to calculate structural indicators
  - Follow-up with paired-sample t-tests for comparison
- MPNet Software for exploring local configurations & the role of peer-support
- Exponential Random Graph (ERG) models: *“By analyzing the characteristics of the two networks in terms of local configurations using Exponential Random Graph models, we can explore and compare the local features of the two networks (p3)”*

# Article 1: Analyses

- Local configurations:  
**Arc, reciprocity, popularity, activity**, transitivity, path closure, and cyclic closure
- ERGMs & Path models (SEM)
- No R Siena

**Table 1** Local configurations and effects of directed networks in ERGMs

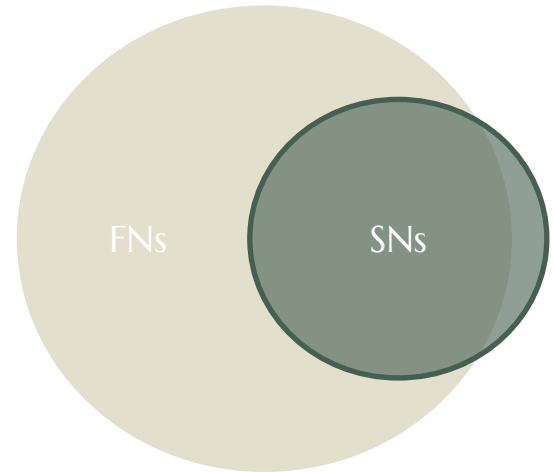
| Name                          | Symbols                                                                             | Meaning of Configurations or Effects                                    | The Meaning of Configurations and Effects in This Study                                                               |
|-------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Local Configuration</b>    |                                                                                     |                                                                         |                                                                                                                       |
| <i>Arc</i>                    |  | The baseline tendency of relationship emergence.                        | The baseline tendency for relationship emergence in FNs or SSNs.                                                      |
| <i>Reciprocity</i>            |  | Tendency for mutual selection.                                          | Two students select each other as friends (or supporters).                                                            |
| <i>Popularity</i>             |  | Tendency for in-degree centrality to be concentrated on certain nodes.  | In a class, some students are chosen by more classmates as friends (or supporters).                                   |
| <i>Activity</i>               |  | Tendency for out-degree centrality to be concentrated on certain nodes. | In a class, some students choose more classmates as friends (or supporters)                                           |
| <i>Triadic transitivity</i>   |  | Transitivity of triadic relationships.                                  | When student i chooses h, and h chooses j as a friend (or supporter), i tends to choose j as a friend (or supporter). |
| <i>Path closure</i>           |  | Path closure is more concentrated in certain areas of the network.      | Tendency for triadic transitivity to occur in clusters based on the relationship between two students.                |
| <i>Cyclic closure</i>         |  | Triadic cycles are more concentrated in certain areas of the network.   | Tendency for triadic cyclicity to occur in clusters based on the relationship between two students.                   |
| <b>Actor relation effects</b> |                                                                                     |                                                                         |                                                                                                                       |
| <i>Sender Effect</i>          |  | People with higher scores on feature X initiate more relationships.     | Students with high perceived peer support tend to choose more friends (or supporters).                                |
| <i>Receiver Effect</i>        |  | People with higher scores on feature X receive more relationships.      | Students with high perceived peer support tend to be chosen by more people as friends (or supporters).                |

1) Each black dot in the table represents an actor, that is, a student in the class; directed lines indicate a directed relationship from the sender to the receiver; 2) The parameter *Arc* is not a direct measurement of network density

# Article 1: (some) Results

- SNs overlap largely with FNs (88.5%)
- Differences between FNs & SNs at the individual, dyadic, triadic, and group level
  - Individual: On average, higher in-degree & out-degree in FNs than SNs
  - Dyadic: More non-reciprocal and reciprocal ties in FNs than SNs
  - Triadic: Fewer empty triads in FNs than SNs
  - Group: Higher average density of FNs

Conclusion: Two distinct network types  
(not interchangeable)



## Who Helps Whom? Investigating the Development of Adolescent Prosocial Relationships

Loes van Rijsewijk, Jan Kornelis Dijkstra, Kim Pattiselanno, Christian Steglich, and René Veenstra  
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We investigated adolescent prosocial relations by examining social networks based on the question “Who helps you (e.g., with homework, with repairing a flat [bicycle] tire, or when you are feeling down)?” The effects of individual characteristics (academic achievement, symptoms of depressive mood, and peer status) on receiving help and giving help were examined, and we investigated the contribution of (dis)similarity between adolescents to the development of prosocial relations. Gender, structural network characteristics, and friendship relations were taken into account. Data were derived from the Social Network Analysis of Risk behavior in Early adolescence (SNARE) study, and contained information on students in 40 secondary school classes across 3 waves ( $N = 840$ ,  $M$  age = 13.4, 49.7% boys). Results from longitudinal social network analyses (RSiena) revealed tendencies toward reciprocation of help and exchange of help within helping groups. Furthermore, boys were less often mentioned as helpers, particularly by girls. Depressed adolescents were less often mentioned as helpers, especially by low-depressed peers. Moreover, lower academic achievers indicated that they received help from their higher achieving peers. Rejected adolescents received help more often, but they less often helped low-rejected peers. Last, low- and high-popular adolescents less often helped each other, and also high-popular adolescents less often helped each other. These findings show that (dis)similarity in these characteristics is an important driving factor underlying the emergence and development of prosocial relations in the peer context, and that prosocial behavior should be defined in terms of benefiting *particular* others.

**Keywords:** prosocial behavior, helping, adolescent, peer, social networks, RSiena

Helping behavior is part of prosocial behavior, which has been defined as “voluntary behavior that benefits others or promotes harmonious relations with others” (e.g., providing emotional or practical help; Dovidio, Piliavin, Schroeder, & Penner, 2006; Eisenberg et al., 1999). Giving and receiving help become salient already in the very early stages of life: Young children tend to respond prosocially to parental or peer distress, are willing to share objects, and comfort upset others (Eisenberg, Fabes, & Spinrad, 2006). As regards receiving help, children depend mainly on their parents, who take up a central role in the provision of practical and emotional support (Furman & Buhrmester, 1992; Larson & Richards, 1991). During the transition to adolescence, however, the context in which giving and receiving help take place partly shifts from parents to peers: Adolescents seek to achieve a higher degree of autonomy from their parents (Allen & Land, 1999; Berndt, 1982), and gradually spend less time with their parents from late childhood into adolescence (Larson & Richards, 1991). Adolescents spend a substantial portion of their waking hours at school in the presence of peers, diminishing the role of parents as help providers. Indeed, although parents

remain key instrumental help providers, peers become an important addition to adolescents’ social support system (del Valle, Bravo, & López, 2010; Hombrados-Mendieta, Gomez-Jacinto, Dominguez-Fuentes, Garcia-Leiva, & Castro-Trav Tráve, 2012), given their familiarity with the challenges age-mates face (Furman & Buhrmester, 1992) and their day-to-day contact.

This shift in context from parents to peers also influences how giving and receiving help are perceived: Given the importance of peers in shaping adolescents’ behaviors and relationships (Adler & Adler, 2003; Baumeister & Leary, 1995; Ormel, Lindenberg, Steverink, & Verbruggen, 1999), *which peers to give help to and which peers to receive help from* become salient questions at this age. Traditionally, research on adolescent helping in the peer context has overlooked this relational nature of prosociality, and mainly focused on explaining adolescent prosocial tendencies as an individual outcome (for a review, see Eisenberg et al., 2006; see exceptions notwithstanding: Baerveldt, van Duijn, Vermeij, & Hemert, 2004; Lomi, Sijnders, Steglich, & Törlo, 2011). Consequently, we know to some extent who is likely to help others, which peers profit from this help, and what characterizes the development of these peer prosocial relations remain largely unknown.

To shift the focus to receivers of help and prosocial relations among peers, in this study we aimed to answer the question “Who helps whom?” We identified adolescent prosocial relationships with peers (i.e., peer relationships of help giving/receiving) by asking participants to nominate those peers that “. . . help you with problems (e.g., with homework, with repairing a flat [bicycle] tire,

# “Who Helps Whom? Investigating the Development of Adolescent Prosocial Relationships (van Rijsewijk, Dijkstra, Pattiselanno, Steglich & Veenstra, 2016)

## Pro-social networks: Giving & receiving help

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# Article 2: Adolescents' pro-social network

- **Children depend mainly on their parents for practical and emotional support**
  - **Changes during the transition to adolescence, start relying on peers**
- Prior research: prosocial tendencies as an individual outcome
  - But what about the relational nature of prosociality?
- Development of pro-social networks
  - *"Who helps you (e.g., with homework, with repairing a flat [bicycle] tire, or when you are feeling down)?"*
- Individual characteristics (academic achievement, symptoms of depressive mood, peer status, gender), structural network characteristics, friendship relations, and the (dis)similarity between adolescents

# Article 2: (some) RQs & Mechanisms

*Which characteristics predict receiving and giving help?*

- Altruism
- Social standing as a relationship-formation mechanism
- Other: Gender, reciprocity, friendship relations, etc.

*Who helps whom? Influence of (dis)similarity*

- Complementary characteristics (low & high academic achiever)
  - Admitting incompetence is less threatening
- Similarity-attraction approach (low & low academic achiever)
  - Needs are more easily understood & communication runs smoothly
  - Lower chance of being rejected or ridiculed

# Article 2: Data & Analysis

- Social Network Analysis of Risk Behavior in Early adolescence (SNARE)
- Dutch students in 40 secondary school classes across 3 waves
- Longitudinal social network analysis
- Stochastic actor-based statistical models of social network dynamics
  - Rsiena, model parameters via an iterative Monte-Carlo Procedure
  - Alternating until convergence, then updating of model parameters
  - Based on discrepancies between the observed & simulated values
  - *“Under which conditions actors will create, maintain, or dissolve a prosocial relation?”*

# Article 2: (some) Results

- Significant similarity effects (depression, popularity & gender)
  - Driven by an aversion to receiving help from dissimilar peers
- Dissimilarity–repulsion hypothesis rather than similarity–attraction approach

# Research questions



03

# Research questions (Based on FIS data)

- Multidimensionality of friendship:
  - Both 'being friends' & emotional intimacy (discussing problems)
  - Two types of ties (multiplex)
- How do these two networks develop over time, when transitioning from tween, to teen, to young adult?
- Do these developments differ based on gender?
- **To what extent do girls and boys differ in the development & maintenance of emotional support networks? (Or: Are girls more likely to transition their friendship networks into emotional support networks than boys?)**

*Which would answer:*

- *Do boys have fewer emotional support networks than girls, as is often claimed?*
- *(how) do (potential) gender differences develop over the course of puberty?*

# Research questions (Explanations)

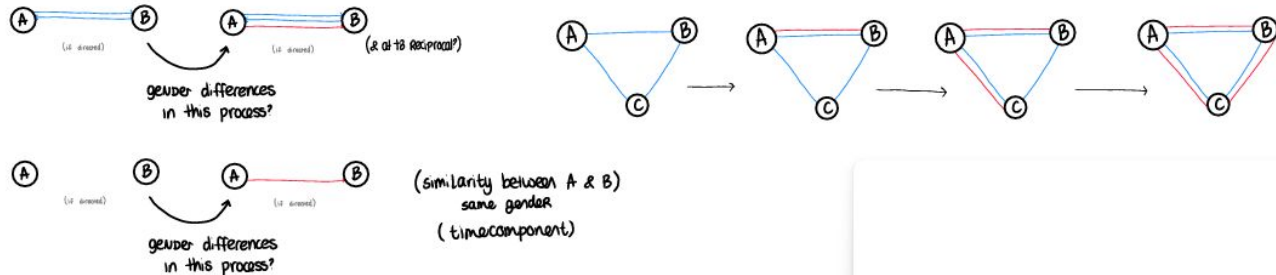
**RQ: Underlying explanation: Homophily, gendered influence processes, or both?**

- Homophily
  - Girls are more likely to self-disclose (Borowski & Rose, 2021)
  - And girls are more likely to befriend girls (van Rijsewijk et al., 2016)
- Gendered influence processes
  - A positive feedback loop
  - Self-disclosing invites self-disclosing (Costello, Bailey, Stern & Allen, 2024)
  - If your friends share with you, you will likely share with them

# Practicalities



**3** How does the overlap come into existence? // when is an emotional intimacy tie formed?



# Research Question (Extra)

*SubRQ: Can a lack of emotional support be “fixed” by entering into a romantic relationship? (How does developing a romantic tie influence the formation & maintenance of an emotional support network?)*

- Scientific evidence underpinning this idea
  - Idea of ‘mankeeping’ (Ferrara & Vergara, 2024)
  - Women ‘opening up’ their network to their partners (emotional labour)

Questions, ideas, feedback?